Supplemental Table S1: Search Strategies in MEDLINE and EMBASE

Database: **OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present**

Search Strategy:

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1 Receptor, ErbB-2/ (22537)

2 HER 2.mp. (7065)

3 human epidermal growth factor receptor 2.mp. (6260)

4 1 or 2 or 3 (28451)

5 cancer*.mp. (1645882)

6 cancer/ (398655)

7 neoplas*.mp. (2746081)

8 neoplasm/ (398655)

9 tumour*.mp. (256440)

10 tumor*.mp. (1794301)

11 malignan*.mp. (537112)

12 carcinoma*.mp. (815840)

13 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 (3865094)

14 4 and 13 (27217)

15 ((HER 2 or HER2 or ErbB2) adj3 (inhibit* or blocker* or antagonist*)).mp. (1923)

16 Antibodies, Monoclonal, Humanized/ (33389)

17 pertuzumab.mp. (889)

18 Perjeta.mp. (21)

19 2C4.mp. (124)

20 15 or 16 or 17 or 18 or 19 (35370)

21 14 and 20 (4856)

22 randomized controlled trial.pt. (480449)

23 controlled clinical trial.pt. (93038)

24 randomized.ab. (440375)

25 placebo.ab. (196992)

26 clinical trials as topic.sh. (186743)

27 randomly.ab. (309248)

28 trial.ti. (197261)

29 22 or 23 or 24 or 25 or 26 or 27 or 28 (1210113)

30 exp animals/ not humans.sh. (4571896)

31 29 not 30 (1112744)

32 21 and 31 (814)

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Database: **Embase <1974 to 2019 March 3>**

Search Strategy:

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1 Human epidermal growth factor receptor/ (3180)

2 HER2.mp. (46588)

3 ErbB-2 receptor.mp. (193)

4 1 or 2 or 3 (47613)

5 cancer/ (77608)

6 cancer.mp. (3205432)

7 neoplas*.mp. (898054)

8 neoplasm/ (410665)

9 carcinoma*.mp. (1096863)

10 tumour*.mp. (348059)

11 tumor*.mp. (2717660)

12 malignan*.mp. (811125)

13 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 (4878417)

14 4 and 13 (46893)

15 Antibodies, Monoclonal, Humanized/ (190493)

16 pertuzumab.mp. (4191)

17 Perjeta.mp. (306)

18 2C4.mp. (173)

19 ((HER 2 or HER2 or ErbB2) adj3 (inhibit* or blocker* or antagonist*)).mp. (3286)

20 15 or 16 or 17 or 18 or 19 (197003)

21 14 and 20 (5992)

22 random:.tw. (1397553)

23 placebo:.mp. (429514)

24 double-blind:.tw. (196573)

25 22 or 23 or 24 (1643579)

26 21 and 25 (934)

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Supplemental Table S2: Risk of bias assessment in the included studies

1- Gianni et al, 2016 (NeoSphere)		
Random sequence generation *(selection bias)*	Low risk	Randomization was done by a central center using dynamic allocation. Eligible patients were randomly assigned using an interactive voice response system and dynamic allocation
Allocation concealment *(selection bias)*	Low risk	As randomization was done by a central center (third party) using an interactive voice response system we can assume allocation code was not broken.
Blinding of participants and personnel *(performance bias)*	High risk	Open-label study
Blinding of outcome assessment *(detection bias)*	Unclear	Authors did not explicitly mention
Incomplete outcome data *(attrition bias)*	High risk	Five patients discontinued treatment in the Pertuzumab group (107) (three from left ventricular dysfunction, one from drug hypersensitivity, and one from abdominal strangulated hernia) and no one stopped the treatment due to side effects in the placebo group.
Selective reporting *(reporting bias)*	Low risk	Predefined according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and results were reported clearly for both arms of the trial.

Risk of bias from other sources	Low risk	Nothing was found
2- Von Minckwitz et al, 2019 (APHINITY)		
Random sequence generation *(selection bias)*	Low risk	Web based system was used for sequence generation which indicated that sequence generation was randomly conducted, minimizing the risk of bias.
Allocation concealment *(selection bias)*	Low risk	No explicit statement about a definite allocation concealment method they used. Authors reported that all parties involved in the trial were unaware of treatment assignment indicated that there is low probability that investigators enrolling participants could foresee assignment of patients to either arm of the trial. Using Permuted randomization decreases the ability to detect further assignment.
Blinding of participants and personnel *(performance bias)*	Low risk	The study is double blinded. Both the participants and the investigators were blinded.
Blinding of outcome assessment *(detection bias)*	Low risk	No explicit mentioning who assessed the outcome and if they were blinded or not but they mentioned that all parties involved were unaware of treatment assignment so we assumed this could include the assessors.

Incomplete outcome data *(attrition bias)*	Low risk	One year of treatment was completed by 84.5% (2028) of the patients in the pertuzumab group and 87.4% (2101) of the patients in the placebo group. Treatment was discontinued for safety reasons by 7.8% (187.2) of the patients in the pertuzumab group and 6.4% (153.92) of the patients in the placebo group. The rate of treatment discontinuation due to adverse events was 1.1 percentage points higher with pertuzumab than with placebo. The adverse-event profile during the treatment period was generally balanced between the two groups.
Selective reporting *(reporting bias)*	Low risk	The outcome was predefined and specified according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and the results were reported. Outcomes were mentioned in the results in both arms of the study in a way that we can use them. They are clear and complete in the tables. The risk of bias for selective reporting is low
Bias from other sources	Low risk	Nothing was found
3- Urruticoechea et al, 2017 (PHEREXA)		
Random sequence generation *(selection bias)*	Low risk	Patients were randomly assigned 1:1 via interactive voice response system and/or interactive web-based response system. Web based response system was used for sequence generation which indicated that sequence generation was randomly conducted, minimizing the risk of bias.
Allocation concealment (selection bias)	Low risk	No explicit statement about a definite allocation concealment method they used. As randomization was done using a third party (interactive web-based….), we assumed that the

		allocation was concealed. There was low probability that investigators enrolling participants could foresee assignment of patients to either arm of the trial.
Blinding of participants and personnel *(performance bias)*	High Risk	Open-label study
Blinding of outcome assessment *(detection bias)*	Low Risk	An independent data monitoring committee monitored safety and made recommendations regarding continuation of the study.
Incomplete outcome data *(attrition bias)*	High risk	From each group, there were 203 participants who withdrew. In the Pertuzumab group, 203 withdrew from 228; 16 for safety issues, and 173 due to disease progression, and 14 for other causes. In the placebo group, 203 withdrew from 218; 19 for safety, 168 from disease progression, 16 for other reasons
Selective reporting *(reporting bias)*	Low risk	Predefined according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and results were reported clearly for both arms of the trial.
Bias from other sources	Low risk	Nothing was found
4-Rimawi et al, 2018 (PERTAIN)		
Random sequence generation *(selection bias)*	Low Risk	Randomization was done by a central center using dynamic allocation. Eligible patients were randomly assigned using an interactive voice response system and dynamic allocation

Allocation concealment *(selection bias)*	Low Risk	As randomization was done by a third party (Almac Clinical Technologies, San Francisco, CA via an interactive voice response system, we can assume that allocation was concealed.
Blinding of participants and personnel *(performance bias)*	High Risk	Open-label study
Blinding of outcome assessment *(detection bias)*	Low Risk	An independent data monitoring committee monitored safety and made recommendations regarding the continuation of the study.
Incomplete outcome data *(attrition bias)*	High risk	Thirteen patients (10.2%) discontinued pertuzumab, as a result of AEs, and 34 patients (26.8%) had pertuzumab interrupted because of AEs'' While no one in the control group discontinued treatment or had interrupted treatment due to side effects.
Selective reporting *(reporting bias)*	Low Risk	Predefined in the methods and reported in the results Predefined according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and results were reported clearly for both arms of the trial.
Risk of bias from other sources	Low Risk	Nothing Found
5-Tabernero et al, 2018 (JACOB)		

Random sequence generation *(selection bias)*	Low risk	Randomization was done by a central permuted block randomization scheme (block size of 4) with an interactive voice or web response system, stratified by geographical region, previous gastrectomy, and HER2 positivity.
Allocation concealment *(selection bias)*	Low risk	The authors mentioned that they performed central randomization and drug pack number allocation using a permuted block randomization scheme (block size of 4) with an interactive voice or web response system (IxRS). They gave a patient number and treatment assignment to the investigator via IxRS at enrolment (the investigator was masked to treatment allocation)
Blinding of participants and personnel *(performance bias)*	Low risk	Double blinded study
Blinding of outcome assessment *(detection bias)*	Low risk	An independent data monitoring committee assessed safety

Incomplete outcome data *(attrition bias)*	Low risk	Adverse events leading to the discontinuation of pertuzumab and trastuzumab or placebo and trastuzumab were reported in 45 (12%) of 385 patients in the pertuzumab group and 45 (12%) of 388 patients in the control group. Serious adverse events that the investigator considered to be related to study treatment were reported in 48 (12%) patients in the pertuzumab group and 40 (10%) patients in the control group. Most deaths were related to disease progression (213 [89%] of 240 in the pertuzumab group and 232 [89%] of 262 in the control group). Adverse events leading to death were reported in 27 (7%) of 385 patients in the pertuzumab group and 30 (8%) of 388 patients in the control group. The withdrawal and the adverse events were balanced between the two arms of the study.
Selective reporting *(reporting bias)*	Low risk	Predefined according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and results were reported clearly for both arms of the trial.
Risk of bias from other sources	Low risk	Nothing was found
6- Shao et al , 2019 (PEONY)		
Random sequence generation *(selection bias)*	Low Risk	Quote: ‘’The study site will obtain the patient randomization number and treatment assignment from the interactive voice/web□based response system (IxRS)’’.

Allocation concealment *(selection bias)*	Low Risk	Quote: ''Patients will be enrolled using a permuted block randomization procedure''
Blinding of participants and personnel *(performance bias)*	Low Risk	Quote: ''randomized, placebo-controlled, double-blind, multicenter clinical trial''.
Blinding of outcome assessment *(detection bias)*	Low Risk	Triple (Participant, Investigator, Outcomes Assessor) blind as stated in the clinicaltrials.gov
Incomplete outcome data *(attrition bias)*	Low Risk	Treatment was completed in 214/219 (97.7%) of the patients in the pertuzumab group and 108/110 (98.2%) of the patients in the placebo group. 210/219 (95.9%) of the patients in the pertuzumab group and 105/110(95.5%) in the placebo group underwent surgery and had a valid tpCR assessment. The treatment discontinuation was balanced between groups.
Selective reporting *(reporting bias)*	Low Risk	Safety data were specified in the protocol and the cardiac safety was described in terms of HF Heart failure (NYHA functional classification III or IV) and Significant LVEF decline events (≥10 percentage points from baseline and to <50%)
Risk of bias from other sources	High risk	Quote: The study was funded by Hoffmann-La Roche Comment: Industry-sponsored

7- Perez et al, 2019 (MARIANNE)		
Random sequence generation *(selection bias)*	Low Risk	Randomization was done by a hierarchical, dynamic random assignment procedure conducted through an interactive voice-response system.
Allocation concealment *(selection bias)*	Low risk	No explicit statement about a definite allocation concealment method they used. However, authors reported that all parties involved in the trial were unaware of treatment assignment indicated that there is low probability that investigators enrolling participants could foresee assignment of patients to either arm of the trial.
Blinding of participants and personnel *(performance bias)*	Low Risk	Blinded with respect to pertuzumab versus placebo.
Blinding of outcome assessment *(detection bias)*	Low Risk	Blinded assessors as mentioned in Clinical trials.gov ''Independent data monitoring and cardiac review committees monitored safety''
Incomplete outcome data *(attrition bias)*	Low Risk	The number of patients who discontinued any treatment component because of AEs was lower in the T-DM1 and T-DM1+pertuzumab arms (T-DM1, 20.8%; TDM1+pertuzumab, 23.0%; HT, 30.6%). The number of patients who died because of AEs was similar across the treatment arms: 5 patients (1.4%) in the T-DM1 arm, and 7 (1.9%) in the TDM1+pertuzumab arm.

Selective reporting *(reporting bias)*	Low Risk	Predefined in the methods and reported in the results. Predefined according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 3.0 and results were reported clearly for both arms of the trial.
Risk of bias from other sources	High Risk	Industry Sponsored, Data were collected by the sponsor and analyzed in collaboration with the authors
8- Swain et al, 2020 (CLEOPATRA)		
Random sequence generation *(selection bias)*	Low risk	Randomization was performed with the use of an interactive voice-response system, with stratification according to geographic region (Asia, Europe, North America, or South America) and prior treatment status (prior adjuvant or neoadjuvant chemotherapy vs. none).
Allocation concealment *(selection bias)*	Low risk	As randomization was done via an interactive voice response system, we can assume that allocation was concealed.
Blinding of participants and personnel *(performance bias)*	Low risk	Double blinded
Blinding of outcome assessment *(detection bias)*	Low risk	Safety was monitored by an independent data and safety monitoring committee and an independent cardiac review committee.

Incomplete outcome data *(attrition bias)*	Low risk	The adverse-event profile during the treatment period was generally balanced between the two groups. The number and causes of deaths other than deaths from disease progression were generally balanced between the two groups, and a similar number of patients died as a result of adverse events (10 [2.5%] in the control group and 8 [2.0%] in the pertuzumab group).
Selective reporting *(reporting bias)*	Low risk	Predefined and reported according to graded according to the National Cancer Institute's Common Terminology Criteria for Adverse Events, version 3.0
Risk of bias from other sources	Low risk	Nothing was found

Supplemental Table S3: Examples of the excluded studies

Study author/year	Reason for exclusion
Walshe J, 2006	Single arm study and not RCT
Portera C, 2008	Single arm study and not RCT
Baselga J, 2010	Single arm study and not RCT
Swain S, 2018 (BERENICE study)	Single arm study and not RCT
Kurzeder 2016	Did not report the outcomes of interest
Martin M, 2016	Did not report the outcomes of interest
Beitsch, 2017 (NBRST study)	Did not report the outcomes of interest
Ian E, 2016	Did not report the outcomes of interest
Beitsch P, 2017	Did not report the outcomes of interest
A.D. Wagner, (INNOVATION study) ClinicalTrials.gov Identifier: NCT02205047	Ongoing
Yamamoto Y, 2018 (PRECIOUS study) UMIN000018202	Ongoing
Kuznar W, 2016 (I-SPY2 trial)	Ongoing. The combination is not fitting the inclusion criteria
Tejal A. Patel, 2019 (TEAL study)	Lapatinib in one arm as a comparator, not placebo.

Schneeweiss A, 2013 (TRYPHAENA study)	Pertuzumab was given in combination of trastuzumab in both arms and the study is comparing different types of chemotherapy
Loibl S, 2017 (GeparSepto study)	Comparing pertuzumab, trastuzumab, chemotherapy versus chemotherapy alone
van Ramshorst M, 2018(TRAIN-2):	Comparing chemotherapy
Kang Y, 2014	The study included Pertuzumab in both arms with two different doses.
Hurvitz S, 2018 (KRISTINE study)	Pertuzumab in both arms
Bachelot T, 2019 (PERUSE study).	Comparing chemotherapy
Makhija 2010	Not all patients were HER2 positive
Kasai H, 2020 (Neo-peaks study) UMIN000014649	Pertuzumab is given in all 3 arms

Supplemental Table S4: GRADE assessment for effect of pertuzumab on asymptomatic/minimally symptomatic LVSD

GRADE domains	Rating	Footnotes	Certainty of the evidence
Risk of Bias *(use the Risk of Bias tables and figures)*	Not serious serious (-1) very serious (-2)	The overall risk of bias across the included studies was not serious. All studies had low risk of bias in the random sequence, allocation concealment domains, and selective reporting of the outcome. Three studies were open label studies and showed high risk of bias regarding blinding of patients and personnel. Three studies had high risk of attrition bias. For the detection bias, only one study the risk was unclear. The two studies of the highest weight did not have high risk of bias in any of the domains. We **did not downgrade** for risk of bias in the included studies.	⨁⨁⨁⨁ High ⨁⨁⨁◯ ~~Moderate~~ ⨁⨁◯◯ Low ⨁◯◯◯ Very Low
Inconsistency	~~Not serious~~ serious (-1) very serious (-2)	By visual inspection of the forest plot of the meta-analysis of the effect of pertuzumab on asymptomatic/minimally symptomatic LVSD, there appeared to be some CIs overlap. Statistical tests showed $Tau^2 = 0.03$; $Chi^2 = 7.40$, df = 6 (P = 0.29); The test of overall effect has p = 0.25 (more than 0.1). We had no evidence to	

		reject the null hypothesis that the studies are similar. I^2 = 19 % which meant that heterogeneity might not be important. We **did not downgrade** for inconsistency.	
Indirectness	Not serious serious (-1) very serious (-2)	Almost all studies matched our pre-specified PICO. Eligible patients in all the studies had HER2-positive breast cancer or other types of cancer as defined in our protocol. The treatment given in all the studies contained pertuzumab in one arm, and placebo in the other arm. The outcome of interest was reported in a similar way across all studies except one study where the decrease in LVEF more than 15% points to less than 50% was considered as a criterion for EF decline. We **did not downgrade** for indirectness.	

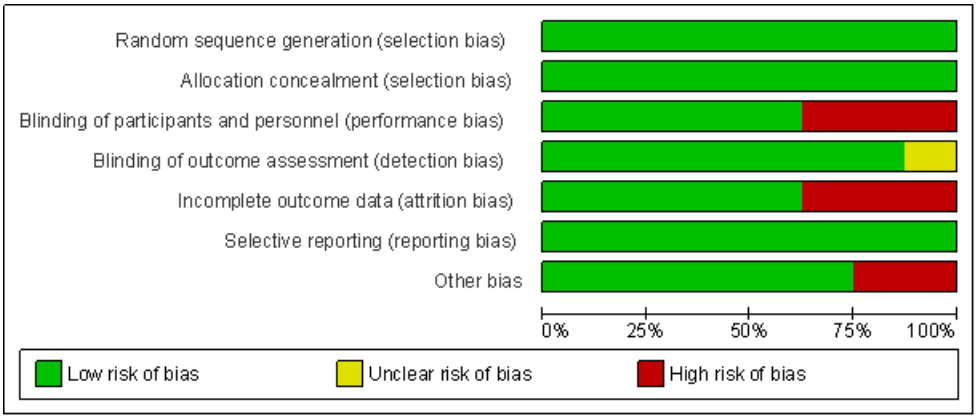
Imprecision	Not serious ~~serious (-1)~~ very serious (-2)	The effect estimate is considered imprecise, as the 95% confidence interval (0.98, 1.61) overlaps with no effect and fails to exclude the clinically important harm. We **rated down** for imprecision.	
Publication Bias	~~Undetected~~ Strongly suspected (-1)	We checked multiple resources (clinical trials conference abstracts, and some of the grey literature sources) to check if all protocols had been published. Publication bias was undetected. We **did not downgrade** for publication bias	
Other (upgrading factors, circle all that apply)	Large effect (+1 or +2) Dose response (+1) Plausible confounding opposing the effect (+1)	N/A since this is an RCT.	

Supplemental Table S5: GRADE assessment for effect of pertuzumab on HF

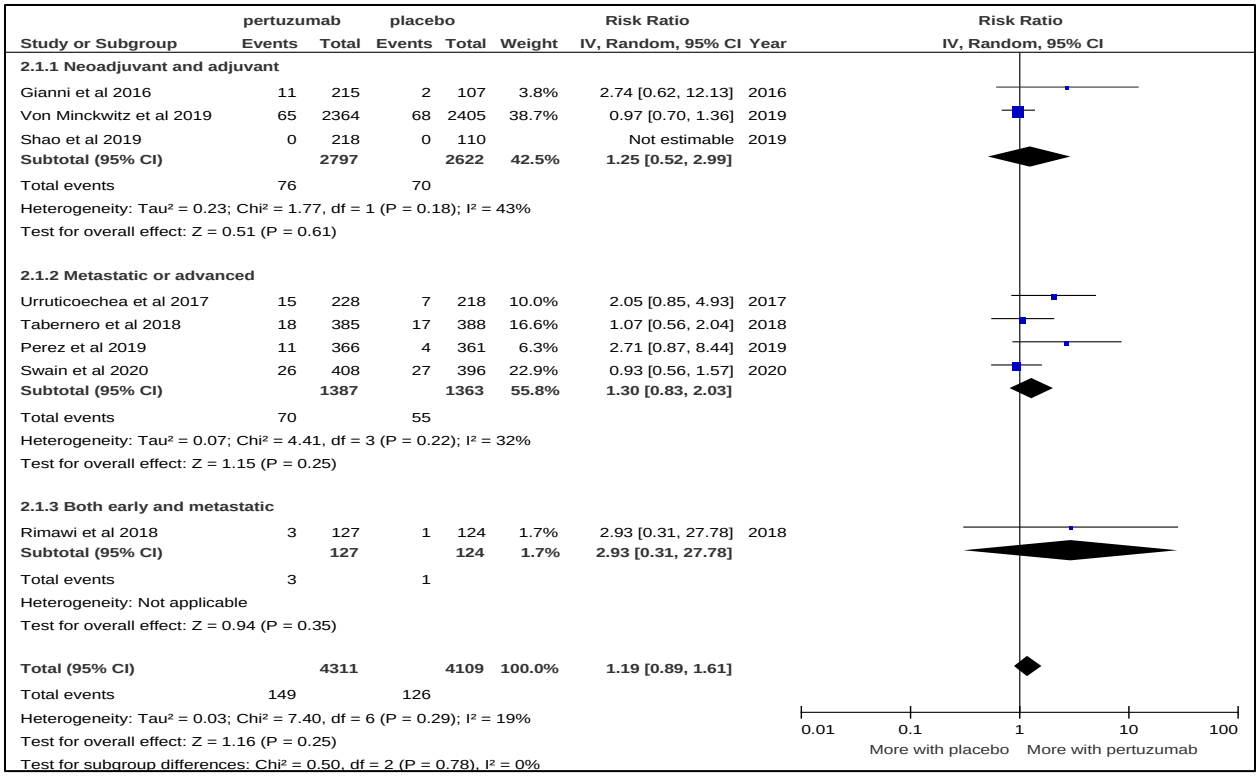
GRADE domains	**Rating**	**Footnotes** (explain reasons for downgrading)	**Quality of the evidence**
Risk of Bias *(use the Risk of Bias tables and figures)*	Not serious serious (-1) very serious (-2)	The overall risk of bias across the included studies was not serious. All studies had low risk of bias in the random sequence, allocation concealment domains, and selective reporting of the outcome. Three studies were open label studies and showed high risk of bias regarding blinding of patients and personnel. Three studies had high risk of attrition bias. For the detection bias, only one study the risk was unclear. The two studies of the highest weight did not have high risk of bias in any of the domains. We **did not downgrade** for risk of bias.	⨁⨁⨁⨁ High ⨁⨁⨁◯ Moderate ⨁⨁◯◯ Low ⨁◯◯◯ Very Low
Inconsistency	Not serious serious (-1) very serious (-2)	By visual inspection of the meta-analysis' forest plot of the effect of pertuzumab on HF, there appeared to be some CIs overlap. Statistical tests showed $Tau^2 = 0$; $Chi^2 = 4.89$, df = 5 (P = 0.43); We had no evidence to reject the null hypothesis that the studies are similar. $I^2 = 0\%$, all of which meant there was no heterogeneity. We **did not downgrade** for inconsistency.	

Indirectness	Not serious serious (-1) very serious (-2)	Almost all studies matched our pre-specified PICO. Eligible patients in all the studies had HER2-positive breast cancer or other types of cancer as defined in our protocol. The treatment given in all studies contained pertuzumab in one arm, and placebo in the other arm. Although the outcome of interest was not reported in a similar way across all studies, we did not think that the effect estimate would have been overestimated, if not underestimated. We **did not downgrade** for indirectness.	

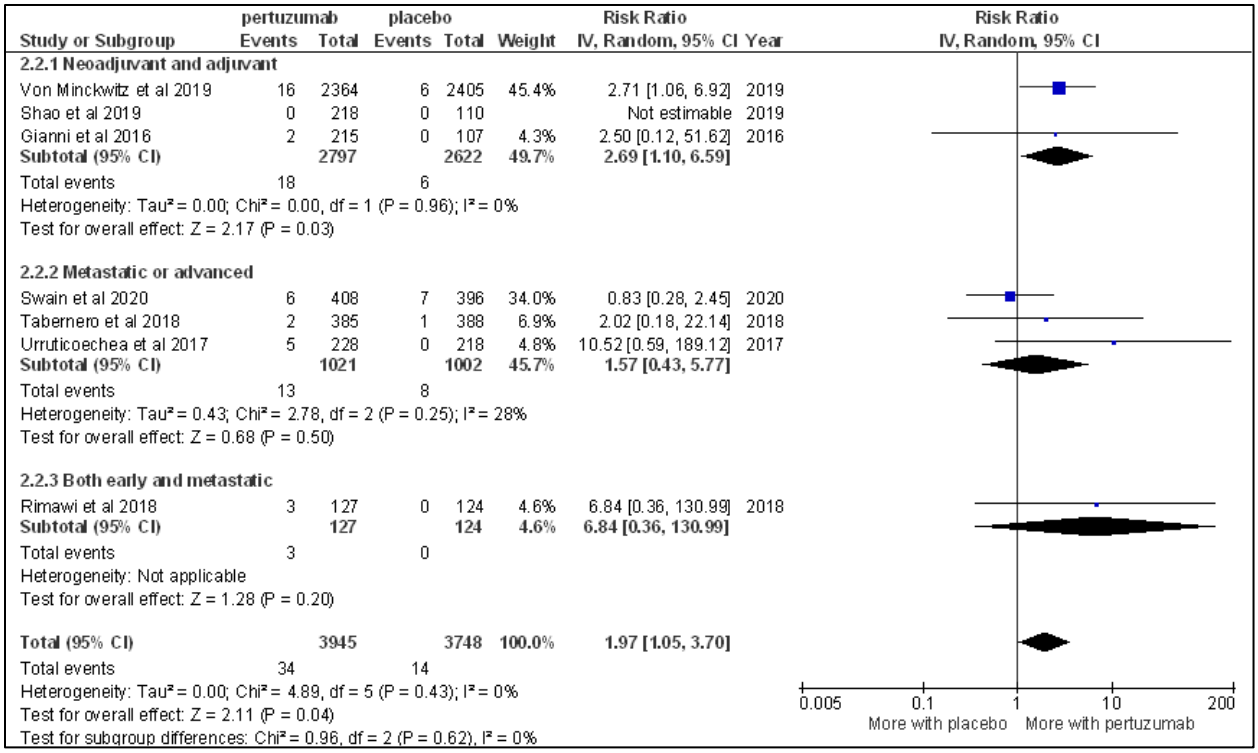
Imprecision	Not serious ~~serious (-1)~~ very serious (-2)	The confidence interval for RR of pertuzumab effect on HF was (1.05 to 3.70). We considered it a wide CI. We also considered the absolute effect: there were 4 more events per 1000 (95%CI, from 0 to 10 more events per 1000). From the literature review, as compared to trastuzumab, we considered the occurrence of 24 events of HF to be an appreciable harmful effect of pertuzumab. The CI of the effect estimate was wide and has not contained a point of appreciable harm. We considered the effect estimate to be imprecise. We **downgraded** for serious imprecision.	
Publication Bias	~~Undetected~~ Strongly suspected (-1)	We checked multiple resources (clinical trials, some of grey literature sources and conference abstracts) to check if all protocols had published their results. Publication bias was undetected. We **did not downgrade**.	
Other (upgrading factors, circle all that apply)	Large effect (+1 or +2) Dose response (+1) Plausible confounding that is opposing the effect (+1)	N/A since this is an RCT.	



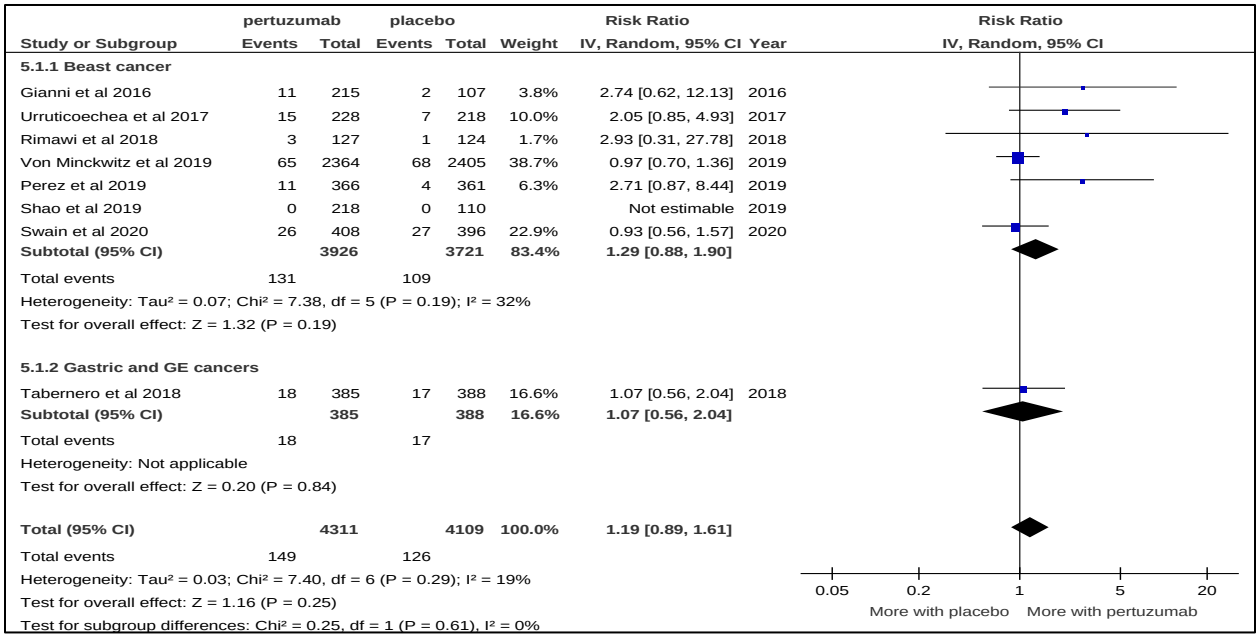
Supplemental Figure S1: Assessment of Risk of Bias by Domain



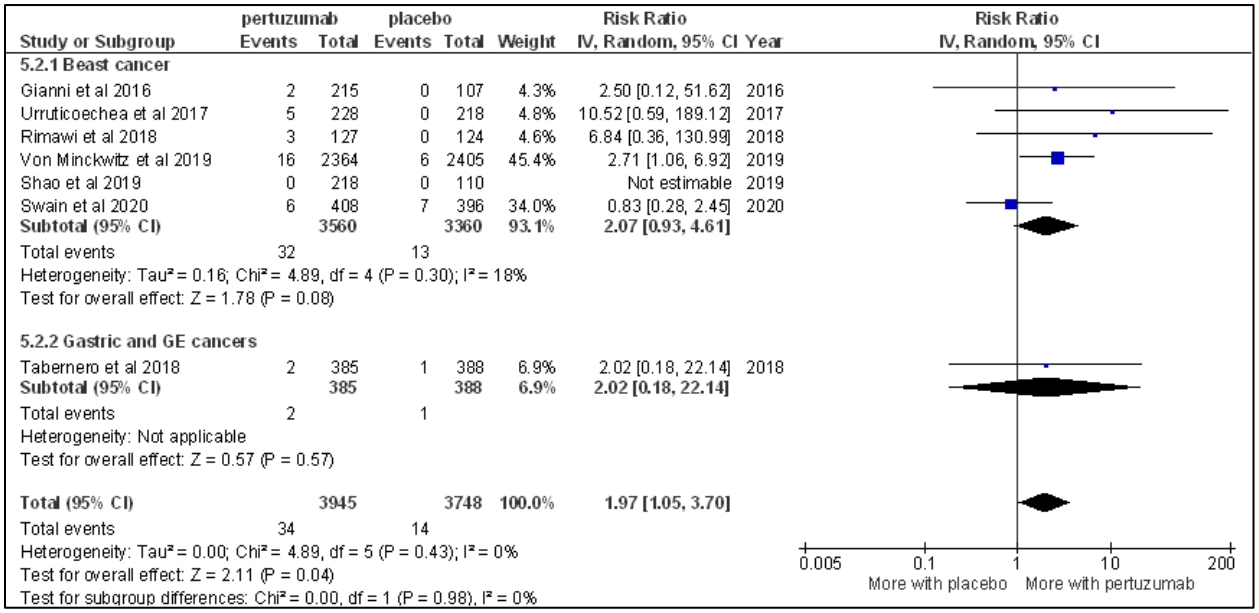
Supplemental Figure S2: Subgroup Analysis of the effect of pertuzumab on asymptomatic LVSD according to treatment setting.



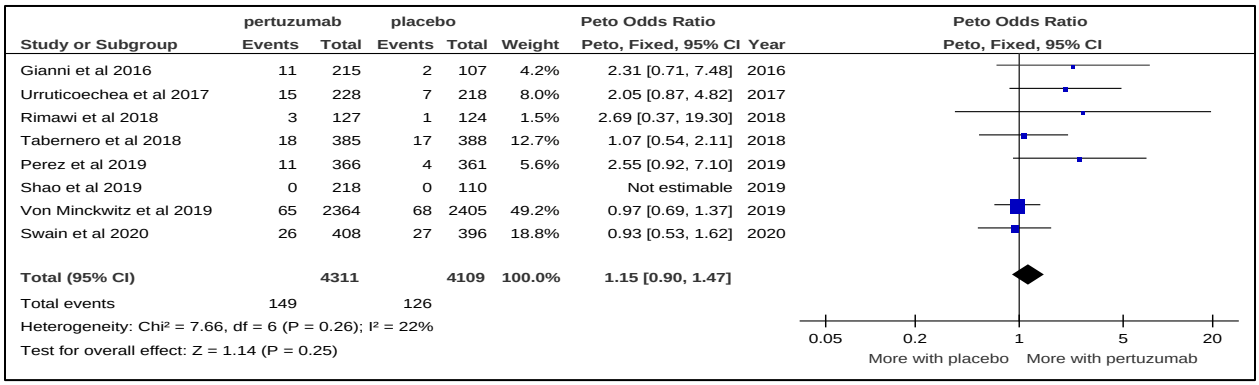
Supplemental Figure S3: Subgroup Analysis of the effect of pertuzumab on HF according to treatment setting



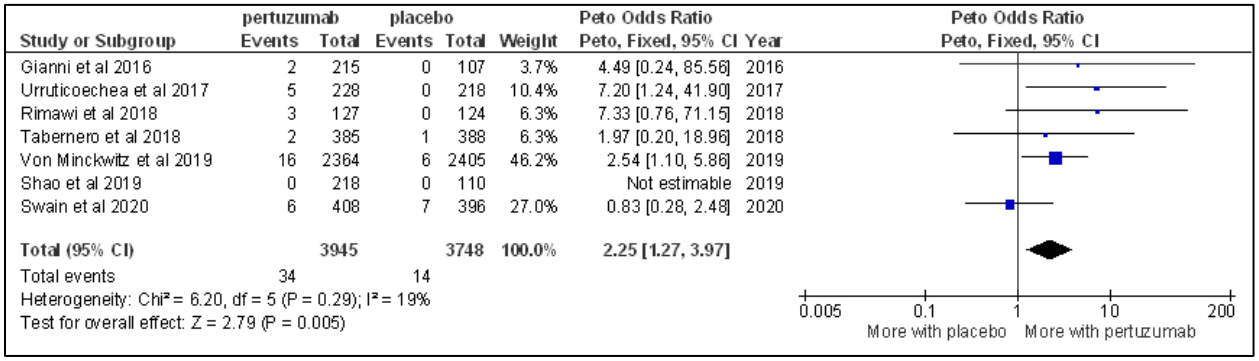
Supplemental Figure S4: Subgroup analysis of the effect of pertuzumab on asymptomatic LVSD according to cancer type



Supplemental Figure S5: Subgroup analysis of the effect of pertuzumab on HF according to cancer type



Supplemental Figure S6: Peto analysis of the effect of pertuzumab on asymptomatic LVSD



Supplemental Figure S7: Peto analysis of the effect of pertuzumab on HF